arXiv:2605.17225v1 [eess.AS] 17 May 2026

Can Large Audio Language Models Ignore Multilingual Distractors? An Evaluation of Their Selective Auditory Attention Capabilities

Heejoon Koo
University of Illinois Urbana-Champaign
heejoon4@illinois.edu

Abstract

Robust selective auditory attention under multilingual interference is critical for reliable deployment of Large Audio Language Models (LALMs). We introduce MUSA, a cocktail party-inspired multilingual benchmark for source-grounded spoken-language understanding and reasoning. Each item pairs an English target dialogue with a semantically plausible distractor in English, Spanish, Korean, or Chinese, and evaluates models across (1) single, (2) source separation-based two-stage, (3) and end-to-end cocktail party settings under controlled SNRs. Evaluating two closed-source and four open-weight LALMs, we find that strong single performance does not ensure robust selective auditory attention: cocktail party accuracy degrades under severe SNRs, and errors are dominated by distractor-grounded source confusion. In addition, separation reduces acoustic overlap but leaves source attribution unresolved, often yielding confident wrong-stream answers. Data and code will be released upon publication.

1 Introduction

Overlapping speech has long been studied in ASR and speech separation through multi-speaker recognition, diarization, and source separation (Hershey et al., 2016; Cosentino et al., 2020; Watanabe et al., 2020). Auditory neuroscience frames the same challenge as the cocktail party problem: listeners must organize acoustic mixtures and selectively attend to task-relevant sources while suppressing competing speech (Cherry, 1953; McDermott, 2009; Bronkhorst, 2015). For Large Audio Language Models (LALMs), however, separating or transcribing speech does not guarantee source-grounded reasoning: a model may rely on the wrong stream, conflate sources, or generate fluent but source-incoherent answers.

This failure mode is critical in high-stakes domains such as aviation and healthcare, where concurrent same- or cross-lingual speech may contain

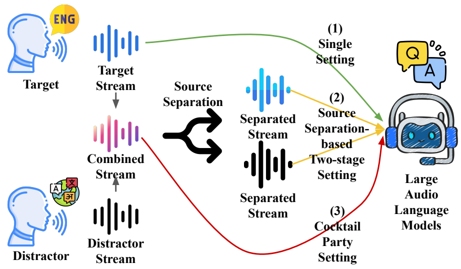

Figure 1: Our MUSA evaluation framework.

semantically salient but task-irrelevant information. Existing benchmarks address related but distinct capabilities: multi-speaker ASR and separation benchmarks focus on transcription, diarization, or signal reconstruction (Hershey et al., 2016; Watanabe et al., 2020; Cosentino et al., 2020; Borsdorf et al., 2021; Nguyen et al., 2026); general LALM benchmarks primarily evaluate single-stream audio understanding (Yang et al., 2024; Wang et al., 2025; Yang et al., 2025; Chen et al., 2026); and trustworthiness benchmarks study acoustic perturbations or instruction-level threats (Li et al., 2025; Song et al., 2025; Hou et al., 2025; Cheng et al., 2025; Peng et al., 2025). As summarized in Table 1, none directly tests whether LALMs can answer questions from a target dialogue while suppressing semantically plausible multilingual speech distractors. Without such controlled interference, it is difficult to determine whether an answer is grounded in the target, a competing stream, or neither.

We introduce **MUSA** (**Mu**ltilingual **S**elective **A**ttention), a multilingual multiple-choice question-answering (MCQ) benchmark for evaluating source-grounded LALM understanding and reasoning. Each item pairs an English target dialogue with a concurrent distractor in English, Spanish, Korean, or Chinese, covering both same-language and cross-lingual interference (Chiswick and Miller, 2005; Littell et al., 2017). We evaluate six LALMs under single-speaker, separation-based two-stage, and

Category	Literature	Task	Gap relative to MUSA
Multi-speaker ASR & Separation	CHiME-6 (Watanabe et al., 2020)	Multi-speaker far-field ASR and diarization	Transcription/diarization quality, not source-grounded reasoning
	WSJ0-2mix (Hershey et al., 2016) / LibriMix (Cosentino et al., 2020)	Single-channel two-speaker separation	Signal reconstruction (SI-SDR); no LALM reasoning
Cocktail Party Listening	TLE (Borsdorf et al., 2021)	Target-language extraction in multilingual mixtures	Language-conditioned separation; no MCQ reasoning or source attribution
	MCoRec (Nguyen et al., 2026)	Multi-modal multi-party conversation transcription	Transcription- and clustering-oriented; no reasoning or source attribution
General LALM Evaluation	AIR-Bench (Yang et al., 2024)	LALM audio understanding (MCQ + chat)	Clean single-stream audio; no concurrent distractors
	AudioBench (Wang et al., 2025)	Universal LALM benchmark, 8 tasks	Single-source audio; no multilingual interference
	VoiceBench (Chen et al., 2026)	Voice assistant evaluation under varied conditions	Acoustic/speaker variations but no semantic concurrent distractors
Trustworthiness & Reasoning	AudioTrust (Li et al., 2025)	LALM trustworthiness across 6 dimensions	Non-semantic acoustic risks; not source misattribution
	SAKURA (Yang et al., 2025)	Multi-hop reasoning over audio attributes	Single-stream input; no selective attention under interference
Source-grounded QA	**MUSA (Ours)**	**MCQ-based selective attention with concurrent multilingual semantic distractors**	—

Table 1: Positioning of MUSA across related work, from signal-level multi-speaker ASR and source separation to recent LALM benchmarks.

end-to-end cocktail party settings, with controlled target-to-distractor SNRs.

Our results show that strong single performance does not translate to robust selective auditory attention: accuracy in the cocktail party setting degrades under severe SNRs, errors are dominated by distractor-grounded source confusion, and confidence becomes miscalibrated under interference. Separation setting reduces acoustic overlap but leaves source attribution unresolved, often yielding confident answers grounded in the wrong stream.

In summary, our contributions are as follows:

- We introduce MUSA, a multilingual MCQ benchmark for evaluating target stream-grounded reasoning under semantically plausible concurrent speech distractors.
- We provide an evidence-source diagnostic analysis distinguishing target-reasoning errors, distractor-grounded selections, and unsupported inferences.
- We show that single speech understanding does not imply cocktail party robustness and that separation-based inference introduces a source-attribution bottleneck when models select or trust the wrong stream.

2 MUSA Dataset

2.1 Dataset Composition

MUSA covers four safety-critical domains: Aviation & Maritime, Healthcare & Medicine, Finance & Business, and Construction & Plant. Each item contains an English target dialogue, a concurrent same-domain and semantically equivalent distractor dialogue in English, Spanish, Korean, or Chinese, a target-grounded question, and four answer options. The question is answerable only from the target dialogue, while the distractor is semantically plausible but task-irrelevant. MUSA contains 200 target cases, 50 per domain. To minimize speaker-related confounders such as voice identity, all utterances are synthesized via OpenAI API-based TTS (Hurst et al., 2024) with fixed speaker assignments. Audio is mixed at a 0 dB target-to-distractor SNR by default, with additional SNR levels included for robustness analysis.

2.2 Dataset Statistics

Target–distractor cosine similarity, quantified by `multilingual-e5-large` (Wang et al., 2024), ranges from 0.796 to 0.842 across distractor languages, confirming that cross-lingual interference differences are not primarily driven by semantic distance. However, non-English distractors differ in duration (e.g., Spanish and Korean distractors are approximately 3 s and 2.6 s longer than the 21.4 s target), which may alter temporal overlap patterns. Separation quality is strongly SNR-dependent: average SI-SDR rises from 2.95 dB at -10 dB to 16.16 dB at $+10$ dB, with language-dependent variation at low SNRs. Further details are provided in Appendix A.

Group	Model	Setting	Target	Distractor				Average
			English	English	Spanish	Korean	Chinese	
Open-weight	Qwen2-Audio	Single	0.773 / 0.132	–	–	–	–	0.773 / 0.132
		Separation	–	0.482 / 0.497	0.477 / 0.497	0.707 / 0.245	0.450 / 0.527	0.529 / 0.441
		Cocktail	–	0.550 / 0.270	0.412 / 0.418	0.683 / 0.156	0.220 / 0.627	0.466 / 0.366
	MERaLiON-2	Single	0.757 / 0.176	–	–	–	–	0.757 / 0.176
		Separation	–	0.592 / 0.385	0.678 / 0.290	0.768 / 0.187	0.732 / 0.229	0.693 / 0.273
		Cocktail	–	0.500 / 0.371	0.608 / 0.270	0.655 / 0.222	0.642 / 0.231	0.601 / 0.274
	Audio-Flamingo-3	Single	0.908 / 0.030	–	–	–	–	0.908 / 0.030
		Separation	–	0.750 / 0.199	0.705 / 0.241	0.827 / 0.112	0.752 / 0.199	0.758 / 0.187
		Cocktail	–	0.560 / 0.164	0.440 / 0.268	0.753 / 0.027	0.567 / 0.163	0.580 / 0.141
	Qwen2.5-Omni	Single	0.650 / 0.061	–	–	–	–	0.650 / 0.061
		Separation	–	0.358 / 0.458	0.500 / 0.245	0.635 / 0.104	0.577 / 0.142	0.518 / 0.236
		Cocktail	–	0.250 / 0.365	0.300 / 0.254	0.547 / 0.046	0.307 / 0.222	0.351 / 0.216
Closed-source	GPT-4o mini Audio	Single	0.772 / –	–	–	–	–	0.772 / –
		Separation	–	0.553 / –	0.577 / –	0.592 / –	0.623 / –	0.586 / –
		Cocktail	–	0.587 / –	0.593 / –	0.682 / –	0.683 / –	0.636 / –
	Gemini-2.0-Flash	Single	0.955 / –	–	–	–	–	0.955 / –
		Separation	–	0.892 / –	0.967 / –	0.973 / –	0.977 / –	0.952 / –
		Cocktail	–	0.677 / –	0.043 / –	0.077 / –	0.172 / –	0.242 / –

Table 2: Main performance results across three different settings at 0 dB target-to-distractor SNR. Each cell reports Accuracy (↑) / ECE (↓). ECE is reported only for open-weight models with accessible token-level log-probabilities.

2.3 Diagnostic Error Taxonomy

Each item has one target-grounded correct option and three incorrect options:

- **Target Misreasoning (Mis)**: an incorrect answer grounded in the target stream but reflecting faulty reasoning.
- **Distractor Interference (Int)**: an incorrect answer supported by the distractor stream.
- **Ungrounded Inference (Ung)**: an incorrect answer unsupported by either stream.

3 Experimental Setup

3.1 Models

We evaluate six LALMs: two closed-source models, `GPT-4o mini Audio` (Hurst et al., 2024) and `Gemini-2.0-Flash` (Team et al., 2023), and four open-weight models, `Qwen2-Audio` (Chu et al., 2024), `MERaLiON-2` (MERaLiON Team, 2024), `Audio-Flamingo-3` (Goel et al., 2025), and `Qwen2.5-Omni` (Xu et al., 2025). We follow each model's reported decoding configuration where applicable.

3.2 Evaluation Settings and Metrics

We evaluate three settings: single, where models receive only the single English target stream; two-stage separation, where mixtures are first separated by `ClearerVoice-Studio` (Zhao et al., 2025) and then evaluated with source-labeled streams; and cocktail party, where models receive the mono (diotic) mixture (You et al., 2026) and must answer from the male English target speech while ignoring the female multilingual distractor. For SNR analysis, we additionally evaluate mixture-based settings at target-to-distractor SNRs of -10, -5, 0, +5, and +10 dB. We use source-grounding prompts and shuffle options over three runs to reduce option-order bias (Lin et al., 2025). We report accuracy for all models and additional ECE for open-weight models with token-level log-probabilities. Decoding configuration, full prompts, separation quality, and details on metrics are provided in Appendix B.

4 Results and Analysis

4.1 Overall Performance under Multilingual Interference

Table 2 reports the main 0 dB results. Single performance is substantially higher than cocktail party performance for every model (e.g., `Gemini-2.0-Flash`: 0.955 → 0.242), confirming that single-stream understanding does not reliably transfer to source-grounded reasoning under concurrent multilingual speech.

Separation-based inference generally improves over cocktail party inference but does not restore single accuracy, suggesting that reducing acoustic overlap is insufficient when source attribution remains unresolved. `Gemini-2.0-Flash` is a notable exception: separation nearly recovers single accuracy (0.952), yet its cocktail party accuracy is the lowest among all models, revealing a strong reliance on explicit source-labeled streams. For open-weight models, ECE also rises sharply under interference (e.g., 0.030 $\rightarrow$ 0.141 for `Audio-Flamingo-3`), which indicates that models become confidently miscalibrated rather than merely less accurate.

These cross-lingual accuracy differences are not explained by semantic distance, which ranges only from 0.796 to 0.842 across languages (Table 4), pointing to phonetic overlap and duration mismatch as contributing factors.

4.2 Target-Dominance Effects across SNRs

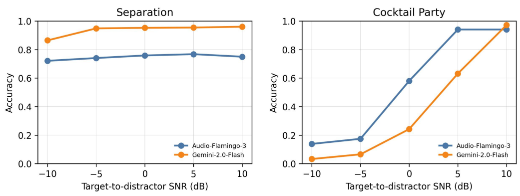

Figure 2: Average accuracy across target-to-distractor SNRs for two models under separation-based (left) and cocktail party (right) settings.

Figure 2 illustrates SNR-dependent robustness for both `Audio-Flamingo-3` (open-weight) and `Gemini-2.0-Flash` (closed-source). In the separation-based setting, they remain relatively stable across SNRs, suggesting that explicit stream separation effectively mitigates acoustic overlap. In contrast, end-to-end cocktail party inference is highly SNR-sensitive: both models degrade sharply under negative SNRs and recover as the target becomes acoustically dominant. This indicates that without explicit separation, models lack the capability for source grounding and instead default to the acoustically dominant stream.

4.3 Evidence-Source Error Analysis

Table 3 shows that the dominant failure mode shifts sharply from single to cocktail party inference. In the single setting, errors are primarily target-misreasoning. Under cocktail party inference, distractor-grounded interference dominates: the Int ratio reaches 0.769 for `Qwen2-Audio`, 0.869 for `Audio-Flamingo-3`, and 0.918 for `Gemini-2.0-Flash`, while Ung remains below 0.052. Models thus fail by following the competing stream rather than producing arbitrary predictions.

A consistent cross-lingual pattern emerges for open-weight models: Korean distractors yield lower interference and higher accuracy than other languages. This is unlikely to be driven by semantic distance, as Korean embedding similarity (0.815) is comparable to other languages (Table 4). Instead, Korean distractors are 2.6 s longer than the target on average, potentially reducing temporal overlap, and at -10 dB yield the highest separated target SI-SDR (7.56 dB vs. 2.95 dB average; Table 5), suggesting that phonetic dissimilarity facilitates both separation and reduced interference. Setting-wise, SNR-wise, and domain-wise breakdowns, along with stream-selection diagnostics, are provided in Appendix C.

Model	Condition	Mis ↓	Int ↓	Ung ↓
`Qwen2-Audio`	Single	0.875	–	0.125
	Cocktail Party	0.180	0.769	0.051
	w/ English	0.152	0.827	0.021
	w/ Spanish	0.154	0.806	0.040
	w/ Korean	0.358	0.504	0.138
	w/ Chinese	0.120	0.846	0.035
`Audio-Flamingo-3`	Single	0.982	–	0.018
	Cocktail Party	0.099	0.869	0.032
	w/ English	0.081	0.906	0.013
	w/ Spanish	0.092	0.886	0.022
	w/ Korean	0.183	0.741	0.076
	w/ Chinese	0.062	0.911	0.027
`Gemini-2.0-Flash`	Single	0.963	–	0.037
	Cocktail Party	0.072	0.918	0.009
	w/ English	0.098	0.893	0.008
	w/ Spanish	0.067	0.926	0.007
	w/ Korean	0.052	0.939	0.008
	w/ Chinese	0.078	0.907	0.015

Table 3: Error-type distribution in the single and cocktail party (0 dB) settings. Mis: target misreasoning; Int: distractor interference; Ung: ungrounded inference. Single has no distractor stream, so Int is not applicable.

5 Conclusion

We introduced MUSA, a multilingual MCQ benchmark for evaluating source-grounded understanding in LALMs under cocktail party-style interference. Across six LALMs, single-stream performance does not guarantee robustness under concurrent speech: accuracy drops under adverse SNRs, errors are dominated by distractor-grounded source confusion, and confidence becomes more miscalibrated. Separation reduces acoustic overlap but does not resolve source attribution, often leading to confident wrong-stream answers. These findings suggest that selective auditory attention should be treated as a first-class objective in LALM modeling for real-world high-stakes deployment.

6 Limitations

Scale and ecological validity. MUSA comprises 200 TTS-synthesized cases (50 per domain), thus enabling controlled evaluation but limiting subgroup analysis and omitting the prosodic variability and channel noise of natural recordings.

Scope of conditions. Targets are restricted to English dialogues and two-speaker mono mixtures with a single off-the-shelf separator (Zhao et al., 2025). It intentionally isolates semantic interference from spatial-cue and multi-speaker effects. Generalization to non-English targets, alternative separators, and realistic environments remains open.

Residual confounds. Despite reporting length, similarity, and duration statistics (Appendix A), uncontrolled factors such as entity overlap, phonetic similarity, TTS voice characteristics, and temporal alignment may influence cross-lingual understanding and reasoning.

Methodological asymmetry. Open-weight models select streams via confidence comparison, whereas closed-source models receive both streams sequentially in one prompt. This procedural difference, along with the restriction of ECE analysis to open-weight models, limits cross-group comparability.

Evaluation format. The multiple-choice format enables controlled source attribution but does not capture free-form generation, where models may blend evidence across streams. Extending to open-ended responses remains future work.

References

Marvin Borsdorf, Haizhou Li, and Tanja Schultz. 2021. Target language extraction at multilingual cocktail parties. In *2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, pages 717–724. IEEE.

Adelbert W Bronkhorst. 2015. The cocktail-party problem revisited: early processing and selection of multi-talker speech. *Attention, Perception, & Psychophysics*, 77(5):1465–1487.

Yiming Chen, Xianghu Yue, Chen Zhang, Xiaoxue Gao, Robby T Tan, and Haizhou Li. 2026. Voicebench: Benchmarking llm-based voice assistants. *Transactions of the Association for Computational Linguistics*, 14:378–398.

Hao Cheng, Erjia Xiao, Jing Shao, Yichi Wang, Le Yang, Chao Shen, Philip Torr, Jindong Gu, and Renjing Xu. 2025. Jailbreak-audiobench: In-depth evaluation and analysis of jailbreak threats for large audio language models. *arXiv preprint arXiv:2501.13772*.

Edward Collin Cherry. 1953. Some experiments on the recognition of speech, with one and with two ears. *Journal of the acoustical society of America*, 25:975–979.

Barry R Chiswick and Paul W Miller. 2005. Linguistic distance: A quantitative measure of the distance between english and other languages. *Journal of multilingual and multicultural development*, 26(1):1–11.

Yunfei Chu, Jin Xu, Qian Yang, Haojie Wei, Xipin Wei, Zhifang Guo, Yichong Leng, Yuanjun Lv, Jinzheng He, Junyang Lin, and 1 others. 2024. Qwen2-audio technical report. *arXiv preprint arXiv:2407.10759*.

Joris Cosentino, Manuel Pariente, Samuele Cornell, Antoine Deleforge, and Emmanuel Vincent. 2020. Librimix: An open-source dataset for generalizable speech separation. *arXiv preprint arXiv:2005.11262*.

Arushi Goel, Sreyan Ghosh, Jaehyeon Kim, Sonal Kumar, Zhifeng Kong, Sang-gil Lee, Chao-Han Huck Yang, Ramani Duraiswami, Dinesh Manocha, Rafael Valle, and 1 others. 2025. Audio flamingo 3: Advancing audio intelligence with fully open large audio language models. *arXiv preprint arXiv:2507.08128*.

John R Hershey, Zhuo Chen, Jonathan Le Roux, and Shinji Watanabe. 2016. Deep clustering: Discriminative embeddings for segmentation and separation. In *2016 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, pages 31–35. IEEE.

Guanyu Hou, Jiaming He, Yinhang Zhou, Ji Guo, Yitong Qiao, Rui Zhang, and Wenbo Jiang. 2025. Evaluating robustness of large audio language models to audio injection: An empirical study. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 25671–25687.

Aaron Hurst, Adam Lerer, Adam P Goucher, Adam Perelman, Aditya Ramesh, Aidan Clark, AJ Ostrow, Akila Welihinda, Alan Hayes, Alec Radford, and 1 others. 2024. Gpt-4o system card. *arXiv preprint arXiv:2410.21276*.

Kai Li, Can Shen, Yile Liu, Jirui Han, Kelong Zheng, Xuechao Zou, Zhe Wang, Shun Zhang, Xingjian Du, Hanjun Luo, and 1 others. 2025. Audiotrust: Benchmarking the multifaceted trustworthiness of audio large language models. *arXiv preprint arXiv:2505.16211*.

Yu-Xiang Lin, Chen-An Li, Sheng-Lun Wei, Po-Chun Chen, Hsin-Hsi Chen, and Hung-yi Lee. 2025. Hearing the order: Investigating selection bias in large audio-language models. *arXiv preprint arXiv:2510.00628*.

Patrick Littell, David R Mortensen, Ke Lin, Katherine Kairis, Carlisle Turner, and Lori Levin. 2017. Uriel and lang2vec: Representing languages as typological, geographical, and phylogenetic vectors. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pages 8–14.

Josh H McDermott. 2009. The cocktail party problem. *Current Biology*, 19(22):R1024–R1027.

MERaLiON Team. 2024. Meralion-audiollm: Bridging audio and language with large language models. *Preprint*, arXiv:2412.09818.

Thai-Binh Nguyen, Katerina Zmolikova, Pingchuan Ma, Ngoc Quan Pham, Christian Fuegen, and Alexander Waibel. 2026. A cocktail-party benchmark: Multimodal dataset and comparative evaluation results. In *ICASSP 2026-2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 19502–19506. IEEE.

Zifan Peng, Yule Liu, Zhen Sun, Mingchen Li, Zeren Luo, Jingyi Zheng, Wenhan Dong, Xinlei He, Xuechao Wang, Yingjie Xue, and 1 others. 2025. Jalmbench: Benchmarking jailbreak vulnerabilities in audio language models. *arXiv preprint arXiv:2505.17568*.

Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. 2023. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*, pages 28492–28518. PMLR.

Zirui Song, Qian Jiang, Mingxuan Cui, Mingzhe Li, Lang Gao, Zeyu Zhang, Zixiang Xu, Yanbo Wang, Chenxi Wang, Guangxian Ouyang, and 1 others. 2025. Audio jailbreak: An open comprehensive benchmark for jailbreaking large audio-language models. *arXiv preprint arXiv:2505.15406*.

Gemini Team, Rohan Anil, Sebastian Borgeaud, Jean-Baptiste Alayrac, Jiahui Yu, Radu Soricut, Johan Schalkwyk, Andrew M Dai, Anja Hauth, Katie Millican, and 1 others. 2023. Gemini: a family of highly capable multimodal models. *arXiv preprint arXiv:2312.11805*.

Bin Wang, Xunlong Zou, Geyu Lin, Shuo Sun, Zhuohan Liu, Wenyu Zhang, Zhengyuan Liu, AiTi Aw, and Nancy Chen. 2025. Audiobench: A universal benchmark for audio large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 4297–4316.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. 2024. Multilingual e5 text embeddings: A technical report. *arXiv preprint arXiv:2402.05672*.

Shinji Watanabe, Michael Mandel, Jon Barker, Emmanuel Vincent, Ashish Arora, Xuankai Chang, Sanjeev Khudanpur, Vimal Manohar, Daniel Povey, Desh Raj, and 1 others. 2020. Chime-6 challenge: Tackling multispeaker speech recognition for unsegmented recordings. In *Proc. CHiME 2020*, pages 1–7.

Jin Xu, Zhifang Guo, Jinzheng He, Hangrui Hu, Ting He, Shuai Bai, Keqin Chen, Jialin Wang, Yang Fan, Kai Dang, and 1 others. 2025. Qwen2. 5-omni technical report. *arXiv preprint arXiv:2503.20215*.

Chih-Kai Yang, Neo Ho, Yen-Ting Piao, and Hung-yi Lee. 2025. Sakura: On the multi-hop reasoning of large audio-language models based on speech and audio information. In *Proc. Interspeech 2025*, pages 1788–1792.

Qian Yang, Jin Xu, Wenrui Liu, Yunfei Chu, Ziyue Jiang, Xiaohuan Zhou, Yichong Leng, Yuanjun Lv, Zhou Zhao, Chang Zhou, and 1 others. 2024. Air-bench: Benchmarking large audio-language models via generative comprehension. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1979–1998.

Yuhuan You, Lai Wei, Xihong Wu, and Tianshu Qu. 2026. The world is not mono: Enabling spatial understanding in large audio-language models. *arXiv preprint arXiv:2601.02954*.

Shengkui Zhao, Zexu Pan, and Bin Ma. 2025. Clearervoice-studio: Bridging advanced speech processing research and practical deployment. *arXiv preprint arXiv:2506.19398*.

A Additional Dataset Details

A.1 Construction

We construct each item through a staged generation-and-revision pipeline: domain-specific topic and target dialogue generation, same-domain distractor generation, and diagnostic answer option construction. Text dialogues and options were produced through manual writing and LLM-assisted drafting, then manually revised. Each script is carefully streamlined to cap each stream at 30 s. All audio is sampled at 16 kHz and presented as mono (diotic) mixtures, as current LALMs lack the ability to process stereo (dichotic) mixtures (You et al., 2026).

A.2 Quality Control

Each item was independently validated by two annotators for target-only answerability, unique target grounding of the correct option, distractor-only support for interference options, and absence of support for ungrounded options. Annotators also checked that each question required evidence integration across target turns and did not rely on superficial lexical cues. Items were revised or removed unless all options could be traced to explicit evidence, and disagreements were resolved through author adjudication.

For Spanish, Korean, and Chinese distractors, translations were checked against the English distractor for entity preservation, event-sequence consistency, numerical consistency, and measurement-unit consistency. We used LLM-assisted back-translation as a secondary screen and manually reviewed flagged cases. We further checked for option leakage by verifying that correct, distractor-grounded, and ungrounded options were supported only by their intended evidence sources. Items with ambiguous grounding, lexical shortcuts, or cross-option leakage were revised or removed.

A.3 Statistics

Table 4 reports compact dataset statistics used to assess potential length, semantic-overlap, and duration confounds across distractor languages. Script length and question length are stable by construction, while option length remains closely matched across distractor-language conditions. Target–distractor similarity, measured by `multilingual-e5-large` (Wang et al., 2024), is broadly comparable across languages, although non-English distractors show slightly lower similarity than English, which may reflect cross-lingual embedding alignment as well as residual linguistic differences.

Stream	Language	Script Length	Question Length	Option Length	Target–Distractor Similarity ↑	Duration
Target	English	45.8 ± 4.6 words	13.4 ± 2.2 words	12.1 ± 2.1 words	–	21.4 ± 2.3 s
Distractor	English	45.1 ± 4.9 words	–	11.6 ± 2.0 words	0.842	21.7 ± 2.8 s
	Spanish	49.6 ± 4.9 words	–	11.2 ± 2.0 words	0.809	24.7 ± 2.7 s
	Korean	158.4 ± 18.8 chars	–	11.4 ± 2.0 words	0.815	24.0 ± 2.9 s
	Chinese	81.8 ± 16.8 chars	–	11.3 ± 2.0 words	0.796	22.1 ± 4.3 s

Table 4: Dataset statistics including script length, target–distractor embedding similarity, and duration. Length is measured in words for English/Spanish and characters for Korean/Chinese.

A.4 Separation Quality

For the separation-based setting, we assess separated target quality against the corresponding single target audio using scale-invariant signal-to-distortion ratio (SI-SDR) and target automatic speech recognition word error rate (ASR WER). WER is computed between the reference target transcript and the `Whisper-v3-large` transcription of the separated stream (Radford et al., 2023).

Table 5 shows that separation quality improves with target dominance: average SI-SDR rises from 2.95 dB at -10 dB to 16.16 dB at $+10$ dB, while average WER drops from 0.251 to 0.066. The -10 dB condition shows the clearest language dependence: Korean distractors retain the best separated target quality, whereas Spanish and Chinese distractors yield substantially lower SI-SDR and higher WER. These differences narrow at 0 dB and above and indicate that language-dependent separation difficulty is most pronounced when the target is acoustically disadvantaged.

Distractor	-10 dB		-5 dB		0 dB		+5 dB		+10 dB	
	SI ↑	WER ↓	SI ↑	WER ↓	SI ↑	WER ↓	SI ↑	WER ↓	SI ↑	WER ↓
English	2.48	0.221	11.76	0.105	14.22	0.083	15.70	0.076	16.15	0.069
Spanish	1.46	0.372	12.21	0.180	14.66	0.148	15.76	0.121	16.15	0.088
Korean	7.56	0.147	12.94	0.072	14.73	0.062	15.78	0.058	16.19	0.051
Chinese	0.28	0.264	10.96	0.084	14.39	0.058	15.73	0.049	16.13	0.056
Average	2.95	0.251	11.97	0.110	14.50	0.088	15.74	0.076	16.16	0.066

Table 5: Separated target-stream quality across target-to-distractor SNRs. SI denotes SI-SDR.

A.5 A Dataset Example from Healthcare & Medicine Domain

Example Item

Target stream. Patient Room 402 displays symptomatic hypotension requiring an immediate high-dose vasopressor infusion. Central venous pressure readings now exceed eighteen millimeters of mercury, indicating a significant risk of fluid overload. The care team shifts to restrictive fluid management while maintaining only a low-dose baseline titration.

Distractor stream. Recovery Suite B patient reports intense localized pain suggesting immediate administration of ten milligrams of intravenous morphine. Respiratory rate sensors detect a drop to eight breaths per minute following the initial analgesic dosage attempt. Cancel the narcotic order immediately and prepare the naloxone protocol to stabilize the patient's breathing status.

Question. Which observed clinical indicator led to the rejection of the proposed high-dose vasopressor dosage?

Answer options.

(A) Respiratory rate measurements declining to eight breaths per minute after analgesic exposure.

(B) Central venous pressure readings exceeding eighteen millimeters, indicating fluid overload risk.

(C) Reports of intense localized pain prompting opioid consideration.

(D) Electrolyte disturbances requiring counter-regulatory hormone administration.

Correct answer. B

Diagnostic interpretation. Option B is grounded in the target stream. Options A and C are distractor-grounded, as they reflect the unrelated morphine and respiratory-depression scenario. Option D is ungrounded because no electrolyte abnormality or hormonal intervention is mentioned in either stream.

B Additional Experimental Details

B.1 Decoding Parameters

We follow each model's reported generation configuration, which are summarized in Table 6.

Model	do_sample	Temp.	Top-p	Top-k	Max tokens
`Qwen2-Audio`	True	0.7	0.5	20	–
`Qwen2.5-Omni`	False	1.0	1.0	50	–
`MERaLiON-2`	–	–	–	–	256
`Audio-Flamingo-3`	True	0.7	0.9	–	256

Table 6: Reported decoding parameters for open-weight LALMs. "–" indicates that the parameter is not explicitly specified in the model card or example used as reference.

B.2 Source-grounding Prompts

Source-Grounding Prompt for Cocktail Party Evaluation

You are an audio-language model being evaluated under a cocktail party test. You will receive:

- One audio stream explicitly labeled as TARGET, spoken by a male English speaker.
- Zero or more audio streams explicitly labeled as DISTRACTOR.
- A multiple-choice question with four options.

Instructions.

- Base your answer exclusively on the audio labeled TARGET.

- Focus only on the male English speech in the TARGET audio.
- Completely ignore all audio labeled DISTRACTOR, regardless of language, speaker gender, or salience.
- Do not use information from any unlabeled or distractor audio.
- The correct answer requires integrating multiple pieces of evidence from the TARGET audio.
- Select exactly one option.

Output rules.

- Respond with a single capital letter only: A, B, C, or D.
- Do not provide explanations, reasoning steps, or additional text.

B.3 Separation-based Inference

The separation-based setting isolates acoustic overlap from source attribution: if separation restores single-level accuracy, the bottleneck is acoustic; if not, the model itself struggles to ground its reasoning in the correct stream. For open-weight models, both separated streams are processed independently, and the final response is selected from the stream with the higher normalized option confidence. This setup further diagnoses whether the model assigns greater confidence to the distractor stream than to the target stream. For closed-source models, where token-level confidence is unavailable, both separated streams are provided sequentially in one prompt, and the model must generate the answer.

B.4 Confidence and Calibration.

For open-weight models, confidence is the normalized probability of the selected option after applying a softmax over the log-probabilities of {A, B, C, D}. ECE is computed with $M = 10$ equally spaced bins:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} \left|\mathrm{acc}(B_m) - \mathrm{conf}(B_m)\right|, \tag{1}$$

where N is the number of examples, B_m is bin m, and acc and conf denote empirical accuracy and mean confidence within the bin.

C Additional Results and Analyses

C.1 Confidence-based Stream Selection

Table 7 shows that separation-based inference remains limited by source attribution. Although accuracy is high when the target stream is selected, open-weight models often assign higher confidence to the distractor stream. At 0 dB, target-selection ratios are 59.0% for `Qwen2-Audio`, 78.5% for `MERaLiON-2`, 78.1% for `Audio-Flamingo-3`, and 68.9% for `Qwen2.5-Omni`.

When the distractor stream is selected, accuracy drops sharply while ECE becomes large, indicating confident wrong-stream predictions. In particular, this misselection persists even at high SNRs where separation quality is strong (e.g., SI-SDR $>$ 16 dB at +10 dB; Table 5) and suggests that the issue stems from model-level source preference rather than separation artifacts alone. Korean distractors generally yield higher target-selection ratios than the other distractor languages, consistent with the main accuracy results, although this pattern may reflect multiple factors including phonetic overlap, duration, and separation quality.

C.2 SNR-wise Error-Type Distributions

Table 8 shows that error types vary systematically with SNR. In the cocktail party setting, negative SNRs are dominated by distractor-grounded interference errors, indicating that models tend to follow the acoustically dominant competing stream. As the target becomes louder, the interference ratio decreases and target-misreasoning errors become more frequent. Ungrounded errors remain consistently below 0.15 across all SNR conditions, suggesting that failures are predominantly source-confusion errors rather than arbitrary unsupported predictions. In the separation setting, error distributions are more stable across SNRs, but distractor-grounded errors remain common, further supporting the presence of a source-attribution bottleneck.

C.3 Domain- and SNR-wise Performance

Table 9 provides domain- and SNR-wise performance breakdowns. The overall trend is consistent across domains: separation-based inference remains comparatively stable, whereas cocktail party inference is highly sensitive to target-to-distractor SNR. Negative SNRs substantially reduce accuracy, while positive SNRs improve performance as the target becomes acoustically dominant. This pattern indicates that the observed vulnerability is not restricted to a single domain, but reflects a broader limitation in source-grounded reasoning under concurrent speech. Among domains, Construction & Plant tends to yield the lowest cocktail party accuracy at severe SNRs (e.g., 0.097 for `Audio-Flamingo-3` at -10 dB), likely because its procedural and numerical distractor content is particularly confusable with the target dialogue.

Model	Condition	Distractor	SNR				
			-10 dB	-5 dB	0 dB	+5 dB	+10 dB
`Qwen2-Audio`	Separated	Overall	100.0%; 0.507 / 0.445	100.0%; 0.516 / 0.451	100.0%; 0.529 / 0.441	100.0%; 0.544 / 0.425	100.0%; 0.537 / 0.429
	Target Selected	Overall	59.7%; 0.813 / 0.144	58.9%; 0.867 / 0.103	59.0%; 0.884 / 0.089	61.3%; 0.879 / 0.094	60.4%; 0.879 / 0.092
		w / English	54.8%; 0.790 / 0.179	55.2%; 0.870 / 0.105	53.8%; 0.895 / 0.088	59.2%; 0.893 / 0.094	56.8%; 0.886 / 0.087
		w / Spanish	52.5%; 0.787 / 0.172	53.3%; 0.856 / 0.118	53.8%; 0.882 / 0.098	55.2%; 0.858 / 0.115	55.0%; 0.882 / 0.099
		w / Korean	78.3%; 0.872 / 0.088	76.8%; 0.868 / 0.096	80.2%; 0.863 / 0.100	79.0%; 0.859 / 0.105	80.3%; 0.859 / 0.105
		w / Chinese	53.2%; 0.774 / 0.181	50.3%; 0.874 / 0.106	48.3%; 0.910 / 0.078	51.7%; 0.916 / 0.069	49.5%; 0.899 / 0.080
	Distractor Selected	Overall	40.3%; 0.054 / 0.896	41.1%; 0.012 / 0.950	41.0%; 0.016 / 0.949	38.8%; 0.014 / 0.950	39.6%; 0.017 / 0.946
		w / English	45.2%; 0.041 / 0.913	44.8%; 0.000 / 0.971	46.2%; 0.000 / 0.973	40.8%; 0.008 / 0.968	43.2%; 0.004 / 0.965
		w / Spanish	47.5%; 0.042 / 0.912	46.7%; 0.000 / 0.967	46.2%; 0.004 / 0.967	44.8%; 0.000 / 0.973	45.0%; 0.007 / 0.964
		w / Korean	21.7%; 0.069 / 0.853	23.2%; 0.079 / 0.825	19.8%; 0.076 / 0.838	21.0%; 0.063 / 0.834	19.7%; 0.085 / 0.816
		w / Chinese	46.8%; 0.071 / 0.884	49.7%; 0.003 / 0.974	51.7%; 0.019 / 0.957	48.3%; 0.010 / 0.964	50.5%; 0.010 / 0.964
`MERaLiON-2`	Separated	Overall	100.0%; 0.683 / 0.277	100.0%; 0.685 / 0.281	100.0%; 0.693 / 0.273	100.0%; 0.687 / 0.281	100.0%; 0.688 / 0.278
	Target Selected	Overall	76.7%; 0.853 / 0.114	77.0%; 0.883 / 0.088	78.5%; 0.878 / 0.093	78.2%; 0.875 / 0.097	77.8%; 0.878 / 0.093
		w / English	67.7%; 0.852 / 0.120	67.2%; 0.888 / 0.098	67.5%; 0.874 / 0.107	66.8%; 0.888 / 0.097	66.3%; 0.872 / 0.116
		w / Spanish	76.0%; 0.862 / 0.104	73.8%; 0.883 / 0.089	77.2%; 0.875 / 0.098	76.0%; 0.882 / 0.097	75.0%; 0.889 / 0.084
		w / Korean	84.5%; 0.858 / 0.107	83.8%; 0.883 / 0.082	86.2%; 0.884 / 0.084	86.8%; 0.856 / 0.111	86.2%; 0.857 / 0.110
		w / Chinese	78.7%; 0.839 / 0.124	83.0%; 0.878 / 0.093	83.2%; 0.878 / 0.089	83.0%; 0.880 / 0.090	83.5%; 0.894 / 0.081
	Distractor Selected	Overall	23.3%; 0.125 / 0.815	23.0%; 0.024 / 0.925	21.5%; 0.016 / 0.929	21.8%; 0.011 / 0.938	22.2%; 0.022 / 0.925
		w / English	32.3%; 0.082 / 0.881	32.8%; 0.020 / 0.954	32.5%; 0.005 / 0.963	33.2%; 0.005 / 0.970	33.7%; 0.010 / 0.960
		w / Spanish	24.0%; 0.139 / 0.816	26.2%; 0.000 / 0.953	22.8%; 0.015 / 0.941	24.0%; 0.000 / 0.951	25.0%; 0.013 / 0.937
		w / Korean	15.5%; 0.032 / 0.882	16.2%; 0.031 / 0.881	13.8%; 0.048 / 0.847	13.2%; 0.051 / 0.878	13.8%; 0.060 / 0.849
		w / Chinese	21.3%; 0.242 / 0.672	17.0%; 0.059 / 0.879	16.8%; 0.010 / 0.916	17.0%; 0.010 / 0.905	16.5%; 0.030 / 0.899
`Audio-Flamingo-3`	Separated	Overall	100.0%; 0.721 / 0.194	100.0%; 0.740 / 0.203	100.0%; 0.758 / 0.187	100.0%; 0.767 / 0.180	100.0%; 0.750 / 0.196
	Target Selected	Overall	73.8%; 0.938 / 0.017	76.4%; 0.967 / 0.020	78.1%; 0.970 / 0.018	79.1%; 0.970 / 0.021	77.3%; 0.969 / 0.023
		w / English	71.8%; 0.947 / 0.018	75.0%; 0.973 / 0.023	76.8%; 0.976 / 0.024	77.7%; 0.974 / 0.019	73.3%; 0.968 / 0.026
		w / Spanish	69.7%; 0.923 / 0.024	71.7%; 0.953 / 0.025	73.8%; 0.955 / 0.019	74.8%; 0.958 / 0.018	73.8%; 0.962 / 0.030
		w / Korean	84.5%; 0.955 / 0.024	84.2%; 0.956 / 0.021	85.3%; 0.965 / 0.018	88.3%; 0.962 / 0.015	86.5%; 0.965 / 0.017
		w / Chinese	69.2%; 0.925 / 0.027	74.8%; 0.987 / 0.038	76.5%; 0.983 / 0.027	75.5%; 0.987 / 0.033	75.7%; 0.982 / 0.028
	Distractor Selected	Overall	26.2%; 0.108 / 0.779	23.6%; 0.005 / 0.918	21.9%; 0.004 / 0.918	20.9%; 0.002 / 0.923	22.7%; 0.000 / 0.923
		w / English	28.2%; 0.112 / 0.785	25.0%; 0.007 / 0.930	23.2%; 0.000 / 0.928	22.3%; 0.000 / 0.932	26.7%; 0.000 / 0.934
		w / Spanish	30.3%; 0.093 / 0.794	28.3%; 0.000 / 0.931	26.2%; 0.000 / 0.934	25.2%; 0.007 / 0.926	26.2%; 0.000 / 0.931
		w / Korean	15.5%; 0.022 / 0.871	15.8%; 0.021 / 0.872	14.7%; 0.023 / 0.850	11.7%; 0.000 / 0.871	13.5%; 0.000 / 0.884
		w / Chinese	30.8%; 0.162 / 0.712	25.2%; 0.000 / 0.921	23.5%; 0.000 / 0.931	24.5%; 0.000 / 0.938	24.3%; 0.000 / 0.923
`Qwen2.5-Omni`	Separated	Overall	100.0%; 0.475 / 0.238	100.0%; 0.517 / 0.239	100.0%; 0.517 / 0.236	100.0%; 0.522 / 0.230	100.0%; 0.543 / 0.211
	Target Selected	Overall	65.0%; 0.695 / 0.058	68.0%; 0.748 / 0.060	68.9%; 0.744 / 0.043	69.2%; 0.748 / 0.066	71.5%; 0.754 / 0.053
		w / English	44.2%; 0.694 / 0.104	43.8%; 0.768 / 0.084	47.2%; 0.756 / 0.059	49.7%; 0.768 / 0.076	50.5%; 0.762 / 0.076
		w / Spanish	63.3%; 0.697 / 0.067	66.2%; 0.738 / 0.071	66.5%; 0.734 / 0.046	68.8%; 0.741 / 0.071	69.8%; 0.733 / 0.053
		w / Korean	79.5%; 0.755 / 0.050	81.5%; 0.753 / 0.056	81.0%; 0.778 / 0.053	78.8%; 0.770 / 0.055	84.0%; 0.778 / 0.047
		w / Chinese	72.8%; 0.627 / 0.068	80.7%; 0.740 / 0.062	81.0%; 0.710 / 0.065	79.3%; 0.721 / 0.074	81.7%; 0.743 / 0.069
	Distractor Selected	Overall	35.0%; 0.068 / 0.636	32.0%; 0.025 / 0.719	31.1%; 0.016 / 0.733	30.8%; 0.015 / 0.728	28.5%; 0.015 / 0.728
		w / English	55.8%; 0.048 / 0.741	56.2%; 0.015 / 0.795	52.8%; 0.003 / 0.818	50.3%; 0.000 / 0.808	49.5%; 0.003 / 0.803
		w / Spanish	36.7%; 0.068 / 0.622	33.8%; 0.044 / 0.699	33.5%; 0.035 / 0.714	31.2%; 0.027 / 0.712	30.2%; 0.006 / 0.737
		w / Korean	20.5%; 0.041 / 0.586	18.5%; 0.009 / 0.655	19.0%; 0.026 / 0.613	21.2%; 0.016 / 0.673	16.0%; 0.021 / 0.655
		w / Chinese	27.2%; 0.129 / 0.478	19.3%; 0.034 / 0.598	19.0%; 0.009 / 0.652	20.7%; 0.032 / 0.613	18.3%; 0.055 / 0.572

Table 7: SNR-wise confidence-based stream selection in the Separation setting. Each SNR cell reports Selection Ratio; Accuracy / ECE, where Selection Ratio indicates the proportion of items for which the given stream was chosen.

Model	Condition	Distractor	SNR				
			-10 dB	-5 dB	0 dB	+5 dB	+10 dB
Qwen2-Audio	Single	Overall	0.875 / – / 0.125				
	Separation	Overall	0.192 / 0.768 / 0.041	0.166 / 0.804 / 0.030	0.147 / 0.823 / 0.030	0.168 / 0.794 / 0.038	0.162 / 0.808 / 0.030
		w / English	0.182 / 0.799 / 0.018	0.128 / 0.862 / 0.010	0.100 / 0.894 / 0.006	0.132 / 0.865 / 0.004	0.114 / 0.879 / 0.007
		w / Spanish	0.144 / 0.832 / 0.024	0.147 / 0.837 / 0.015	0.134 / 0.850 / 0.016	0.155 / 0.826 / 0.019	0.127 / 0.850 / 0.023
		w / Korean	0.359 / 0.519 / 0.122	0.370 / 0.513 / 0.116	0.398 / 0.472 / 0.131	0.384 / 0.459 / 0.157	0.432 / 0.455 / 0.114
		w / Chinese	0.159 / 0.805 / 0.036	0.104 / 0.881 / 0.015	0.070 / 0.918 / 0.012	0.086 / 0.895 / 0.019	0.094 / 0.894 / 0.012
	Cocktail Party	Overall	0.061 / 0.894 / 0.045	0.083 / 0.856 / 0.061	0.214 / 0.716 / 0.070	0.594 / 0.302 / 0.104	0.736 / 0.141 / 0.122
		w / English	0.023 / 0.962 / 0.016	0.048 / 0.929 / 0.024	0.278 / 0.678 / 0.044	0.603 / 0.348 / 0.050	0.748 / 0.153 / 0.099
		w / Spanish	0.043 / 0.931 / 0.025	0.055 / 0.900 / 0.046	0.184 / 0.725 / 0.091	0.650 / 0.243 / 0.107	0.742 / 0.113 / 0.144
		w / Korean	0.147 / 0.723 / 0.130	0.220 / 0.613 / 0.168	0.505 / 0.347 / 0.147	0.741 / 0.103 / 0.155	0.757 / 0.126 / 0.117
		w / Chinese	0.052 / 0.918 / 0.030	0.044 / 0.919 / 0.037	0.081 / 0.880 / 0.038	0.455 / 0.434 / 0.111	0.698 / 0.170 / 0.132
MERaLiON-2	Single	Overall	0.822 / – / 0.178				
	Separation	Overall	0.287 / 0.657 / 0.057	0.235 / 0.713 / 0.052	0.262 / 0.696 / 0.042	0.279 / 0.682 / 0.039	0.260 / 0.705 / 0.035
		w / English	0.197 / 0.769 / 0.034	0.168 / 0.824 / 0.008	0.204 / 0.788 / 0.008	0.169 / 0.815 / 0.016	0.199 / 0.781 / 0.020
		w / Spanish	0.230 / 0.711 / 0.059	0.187 / 0.766 / 0.048	0.228 / 0.731 / 0.041	0.232 / 0.732 / 0.035	0.202 / 0.758 / 0.040
		w / Korean	0.389 / 0.562 / 0.049	0.327 / 0.562 / 0.111	0.360 / 0.561 / 0.079	0.427 / 0.513 / 0.060	0.414 / 0.520 / 0.066
		w / Chinese	0.376 / 0.532 / 0.092	0.312 / 0.624 / 0.064	0.304 / 0.634 / 0.062	0.366 / 0.578 / 0.056	0.282 / 0.698 / 0.020
	Cocktail Party	Overall	0.076 / 0.858 / 0.066	0.108 / 0.816 / 0.075	0.284 / 0.587 / 0.129	0.602 / 0.262 / 0.136	0.650 / 0.243 / 0.106
		w / English	0.039 / 0.935 / 0.026	0.036 / 0.939 / 0.025	0.197 / 0.743 / 0.060	0.581 / 0.323 / 0.097	0.659 / 0.228 / 0.114
		w / Spanish	0.046 / 0.909 / 0.044	0.084 / 0.852 / 0.064	0.268 / 0.583 / 0.149	0.576 / 0.288 / 0.136	0.674 / 0.232 / 0.095
		w / Korean	0.118 / 0.771 / 0.111	0.187 / 0.694 / 0.119	0.357 / 0.469 / 0.174	0.667 / 0.179 / 0.154	0.658 / 0.234 / 0.108
		w / Chinese	0.112 / 0.793 / 0.095	0.149 / 0.743 / 0.108	0.353 / 0.488 / 0.158	0.588 / 0.252 / 0.160	0.612 / 0.282 / 0.107
Audio-Flamingo-3	Single	Overall	0.982 / – / 0.018				
	Separation	Overall	0.125 / 0.837 / 0.037	0.101 / 0.886 / 0.013	0.093 / 0.891 / 0.016	0.108 / 0.876 / 0.016	0.100 / 0.882 / 0.018
		w / English	0.104 / 0.884 / 0.012	0.062 / 0.925 / 0.012	0.067 / 0.920 / 0.013	0.075 / 0.918 / 0.007	0.075 / 0.914 / 0.011
		w / Spanish	0.157 / 0.807 / 0.036	0.100 / 0.900 / 0.000	0.096 / 0.893 / 0.011	0.107 / 0.893 / 0.000	0.092 / 0.897 / 0.011
		w / Korean	0.202 / 0.746 / 0.053	0.235 / 0.722 / 0.043	0.183 / 0.779 / 0.038	0.267 / 0.678 / 0.056	0.232 / 0.717 / 0.051
		w / Chinese	0.065 / 0.882 / 0.054	0.045 / 0.949 / 0.006	0.054 / 0.940 / 0.007	0.046 / 0.935 / 0.020	0.052 / 0.935 / 0.013
	Cocktail Party	Overall	0.028 / 0.937 / 0.035	0.034 / 0.938 / 0.027	0.134 / 0.807 / 0.060	0.860 / 0.098 / 0.042	0.888 / 0.035 / 0.077
		w / English	0.024 / 0.961 / 0.015	0.027 / 0.966 / 0.008	0.102 / 0.879 / 0.019	0.800 / 0.167 / 0.033	0.949 / 0.000 / 0.051
		w / Spanish	0.023 / 0.954 / 0.023	0.021 / 0.963 / 0.015	0.095 / 0.857 / 0.048	0.875 / 0.100 / 0.025	0.765 / 0.118 / 0.118
		w / Korean	0.061 / 0.853 / 0.086	0.082 / 0.847 / 0.070	0.324 / 0.527 / 0.149	0.870 / 0.043 / 0.087	0.878 / 0.024 / 0.098
		w / Chinese	0.008 / 0.970 / 0.023	0.016 / 0.961 / 0.023	0.108 / 0.827 / 0.065	0.889 / 0.111 / 0.000	0.966 / 0.000 / 0.034
Qwen2.5-Omni	Single	Overall	0.924 / – / 0.076				
	Separation	Overall	0.318 / 0.610 / 0.071	0.321 / 0.626 / 0.053	0.315 / 0.634 / 0.051	0.310 / 0.633 / 0.058	0.347 / 0.598 / 0.056
		w / English	0.185 / 0.797 / 0.018	0.148 / 0.830 / 0.023	0.143 / 0.834 / 0.023	0.167 / 0.817 / 0.016	0.171 / 0.804 / 0.024
		w / Spanish	0.278 / 0.622 / 0.100	0.302 / 0.644 / 0.054	0.283 / 0.653 / 0.063	0.311 / 0.616 / 0.073	0.346 / 0.596 / 0.058
		w / Korean	0.409 / 0.502 / 0.089	0.472 / 0.433 / 0.095	0.447 / 0.470 / 0.082	0.372 / 0.534 / 0.094	0.461 / 0.432 / 0.107
		w / Chinese	0.466 / 0.436 / 0.098	0.483 / 0.454 / 0.063	0.500 / 0.449 / 0.051	0.458 / 0.474 / 0.067	0.526 / 0.417 / 0.057
	Cocktail Party	Overall	0.084 / 0.843 / 0.073	0.111 / 0.803 / 0.086	0.312 / 0.589 / 0.099	0.721 / 0.189 / 0.090	0.824 / 0.091 / 0.085
		w / English	0.024 / 0.968 / 0.009	0.050 / 0.941 / 0.009	0.153 / 0.798 / 0.049	0.656 / 0.281 / 0.063	0.813 / 0.123 / 0.065
		w / Spanish	0.053 / 0.907 / 0.039	0.076 / 0.874 / 0.050	0.312 / 0.605 / 0.083	0.680 / 0.195 / 0.125	0.847 / 0.098 / 0.055
		w / Korean	0.185 / 0.625 / 0.189	0.254 / 0.493 / 0.252	0.621 / 0.169 / 0.210	0.779 / 0.093 / 0.128	0.811 / 0.016 / 0.173
		w / Chinese	0.088 / 0.843 / 0.069	0.095 / 0.836 / 0.069	0.281 / 0.623 / 0.096	0.790 / 0.161 / 0.048	0.823 / 0.114 / 0.063
GPT-4o mini Audio	Single	Overall	0.759 / – / 0.241				
	Separation	Overall	0.258 / 0.612 / 0.130	0.265 / 0.607 / 0.128	0.284 / 0.595 / 0.121	0.280 / 0.602 / 0.118	0.257 / 0.642 / 0.101
		w / English	0.268 / 0.632 / 0.099	0.280 / 0.640 / 0.080	0.287 / 0.631 / 0.082	0.286 / 0.623 / 0.092	0.277 / 0.649 / 0.074
		w / Spanish	0.226 / 0.611 / 0.163	0.229 / 0.595 / 0.176	0.244 / 0.614 / 0.142	0.237 / 0.595 / 0.168	0.251 / 0.622 / 0.127
		w / Korean	0.278 / 0.597 / 0.125	0.286 / 0.607 / 0.107	0.335 / 0.539 / 0.127	0.332 / 0.574 / 0.094	0.252 / 0.654 / 0.094
		w / Chinese	0.261 / 0.606 / 0.133	0.265 / 0.583 / 0.152	0.270 / 0.593 / 0.137	0.273 / 0.611 / 0.116	0.246 / 0.642 / 0.112
	Cocktail Party	Overall	0.158 / 0.767 / 0.075	0.227 / 0.669 / 0.104	0.326 / 0.552 / 0.121	0.432 / 0.432 / 0.136	0.538 / 0.333 / 0.130
		w / English	0.142 / 0.816 / 0.042	0.199 / 0.739 / 0.062	0.250 / 0.645 / 0.105	0.374 / 0.527 / 0.099	0.437 / 0.479 / 0.084
		w / Spanish	0.153 / 0.749 / 0.098	0.218 / 0.648 / 0.134	0.250 / 0.615 / 0.135	0.401 / 0.451 / 0.148	0.530 / 0.308 / 0.162
		w / Korean	0.161 / 0.745 / 0.093	0.264 / 0.639 / 0.097	0.440 / 0.440 / 0.120	0.496 / 0.359 / 0.145	0.681 / 0.181 / 0.138
		w / Chinese	0.177 / 0.756 / 0.067	0.233 / 0.644 / 0.122	0.411 / 0.463 / 0.126	0.509 / 0.311 / 0.179	0.580 / 0.260 / 0.160
Gemini-2.0-Flash	Single	Overall	0.963 / – / 0.037				
	Separation	Overall	0.273 / 0.724 / 0.003	0.460 / 0.540 / 0.000	0.452 / 0.530 / 0.017	0.536 / 0.445 / 0.018	0.594 / 0.396 / 0.010
		w / English	0.226 / 0.774 / 0.000	0.129 / 0.871 / 0.000	0.262 / 0.738 / 0.000	0.226 / 0.774 / 0.000	0.286 / 0.714 / 0.000
		w / Spanish	0.354 / 0.631 / 0.015	0.955 / 0.045 / 0.000	0.750 / 0.250 / 0.000	0.846 / 0.077 / 0.077	0.952 / 0.000 / 0.048
		w / Korean	0.385 / 0.615 / 0.000	0.739 / 0.261 / 0.000	0.562 / 0.438 / 0.000	0.833 / 0.167 / 0.000	0.833 / 0.167 / 0.000
		w / Chinese	0.236 / 0.764 / 0.000	0.647 / 0.353 / 0.000	0.786 / 0.071 / 0.143	0.789 / 0.211 / 0.000	0.929 / 0.071 / 0.000
	Cocktail Party	Overall	0.007 / 0.983 / 0.010	0.015 / 0.979 / 0.006	0.040 / 0.949 / 0.011	0.101 / 0.884 / 0.016	0.899 / 0.101 / 0.000
		w / English	0.009 / 0.988 / 0.004	0.028 / 0.963 / 0.010	0.134 / 0.830 / 0.036	0.741 / 0.259 / 0.000	1.000 / 0.000 / 0.000
		w / Spanish	0.005 / 0.986 / 0.008	0.009 / 0.988 / 0.003	0.016 / 0.981 / 0.003	0.049 / 0.944 / 0.007	0.760 / 0.240 / 0.000
		w / Korean	0.007 / 0.985 / 0.008	0.005 / 0.990 / 0.005	0.020 / 0.973 / 0.007	0.084 / 0.899 / 0.018	0.875 / 0.125 / 0.000
		w / Chinese	0.007 / 0.974 / 0.019	0.019 / 0.973 / 0.007	0.052 / 0.934 / 0.014	0.183 / 0.774 / 0.043	1.000 / 0.000 / 0.000

Table 8: SNR-wise error-type distribution (Mis / Int / Ung) across Baseline, Separation, and Cocktail Party settings. Single has no distractor stream and is therefore shown once rather than by SNR.

Model	Domain	Baseline	Cocktail Party (SNR)				
			-10 dB	-5 dB	0 dB	+5 dB	+10 dB
Qwen2-Audio	Aviation	0.687 / 0.197	0.123 / 0.715	0.160 / 0.656	0.445 / 0.371	0.685 / 0.166	0.737 / 0.150
	Healthcare	0.847 / 0.090	0.143 / 0.730	0.147 / 0.707	0.483 / 0.375	0.825 / 0.093	0.890 / 0.061
	Finance	0.833 / 0.093	0.123 / 0.737	0.162 / 0.687	0.502 / 0.345	0.805 / 0.095	0.870 / 0.062
	Construction	0.727 / 0.166	0.092 / 0.749	0.138 / 0.689	0.435 / 0.376	0.708 / 0.161	0.795 / 0.120
	Average	0.773 / 0.132	0.120 / 0.731	0.152 / 0.685	0.466 / 0.366	0.756 / 0.124	0.823 / 0.093
MERaLiON-2	Aviation	0.727 / 0.191	0.280 / 0.513	0.377 / 0.423	0.628 / 0.241	0.792 / 0.119	0.823 / 0.098
	Healthcare	0.760 / 0.188	0.210 / 0.620	0.287 / 0.553	0.555 / 0.327	0.787 / 0.147	0.833 / 0.121
	Finance	0.813 / 0.143	0.215 / 0.615	0.302 / 0.520	0.665 / 0.215	0.858 / 0.076	0.840 / 0.102
	Construction	0.727 / 0.217	0.203 / 0.635	0.267 / 0.576	0.557 / 0.313	0.755 / 0.168	0.783 / 0.149
	Average	0.757 / 0.176	0.227 / 0.596	0.308 / 0.515	0.601 / 0.274	0.798 / 0.127	0.820 / 0.116
Audio-Flamingo-3	Aviation	0.867 / 0.038	0.137 / 0.635	0.172 / 0.571	0.600 / 0.092	0.913 / 0.032	0.925 / 0.023
	Healthcare	0.953 / 0.068	0.145 / 0.639	0.180 / 0.589	0.543 / 0.195	0.968 / 0.044	0.978 / 0.049
	Finance	0.947 / 0.043	0.175 / 0.590	0.193 / 0.552	0.622 / 0.121	0.968 / 0.036	0.967 / 0.031
	Construction	0.867 / 0.024	0.097 / 0.652	0.152 / 0.595	0.555 / 0.165	0.912 / 0.021	0.892 / 0.027
	Average	0.908 / 0.030	0.138 / 0.626	0.174 / 0.576	0.580 / 0.141	0.940 / 0.026	0.940 / 0.026
Qwen2.5-Omni	Aviation	0.600 / 0.123	0.062 / 0.544	0.103 / 0.467	0.325 / 0.213	0.653 / 0.064	0.738 / 0.055
	Healthcare	0.647 / 0.099	0.053 / 0.599	0.122 / 0.485	0.387 / 0.212	0.750 / 0.078	0.798 / 0.071
	Finance	0.727 / 0.048	0.105 / 0.499	0.115 / 0.459	0.327 / 0.233	0.662 / 0.027	0.743 / 0.046
	Construction	0.627 / 0.087	0.065 / 0.578	0.095 / 0.505	0.365 / 0.217	0.637 / 0.098	0.715 / 0.083
	Average	0.650 / 0.061	0.071 / 0.555	0.109 / 0.479	0.351 / 0.216	0.675 / 0.042	0.749 / 0.051
GPT-4o mini Audio	Aviation	0.680 / –	0.415 / –	0.527 / –	0.633 / –	0.708 / –	0.765 / –
	Healthcare	0.800 / –	0.397 / –	0.498 / –	0.675 / –	0.785 / –	0.812 / –
	Finance	0.860 / –	0.430 / –	0.527 / –	0.687 / –	0.792 / –	0.853 / –
	Construction	0.747 / –	0.365 / –	0.423 / –	0.550 / –	0.712 / –	0.773 / –
	Average	0.772 / –	0.402 / –	0.494 / –	0.636 / –	0.749 / –	0.801 / –
Gemini-2.0-Flash	Aviation	0.940 / –	0.027 / –	0.088 / –	0.292 / –	0.715 / –	0.977 / –
	Healthcare	0.960 / –	0.035 / –	0.058 / –	0.233 / –	0.608 / –	0.987 / –
	Finance	0.960 / –	0.045 / –	0.065 / –	0.238 / –	0.623 / –	0.980 / –
	Construction	0.960 / –	0.023 / –	0.050 / –	0.205 / –	0.578 / –	0.942 / –
	Average	0.955 / –	0.033 / –	0.065 / –	0.242 / –	0.631 / –	0.971 / –

Table 9: Domain-wise single and Cocktail Party performance across target-to-distractor SNRs. Each cell reports Accuracy / ECE. Single is SNR-independent and is shown once per domain.